RESEARCH BRIEF

Can OCR and AI vision reconstruct diagrams reliably?

Ten Finnish-context figures with known source graphs | 83 nodes | 113 edges | 206 printed text regions

Finding

Blind vision reconstructed only 4 of 10 complete figures without a scored error.

OCR read text well but did not infer connections or arrows; the evidence-only OCR union did not change the vision graph.

Approach matrix

Each rate uses 206 source items. OCR rows score printed text; the vision row scores complete graph elements.

Approach	Scored	Errors / 206	Error-free	Key result
RapidOCR PP-OCRv6 Finnish	Text	**18 (8.7%)**	4 / 10	203/206 exact; 15 extra boxes
PaddleOCR PP-OCRv5 Finnish	Text	**59 (28.6%)**	0 / 10	187/206 exact; 40 extra boxes
Older RapidOCR baseline	Text	**61 (29.6%)**	0 / 10	145/206 exact; 0 extra boxes
Blind Codex vision	Graph	**26 (12.6%)**	4 / 10	101/113 edges fully correct

What the graph errors mean

Nodes found: 83/83. Connections found: 105/113. Fully correct edges: 101/113.

The vision graph also had 4 extra edges, 10/83 shape errors, 10/113 edge-label errors, 10/113 arrow errors or omissions, and 8/113 dashed-versus-solid stroke errors.

These categories overlap. A fully correct edge must have the right endpoints, arrow, printed label and stroke.

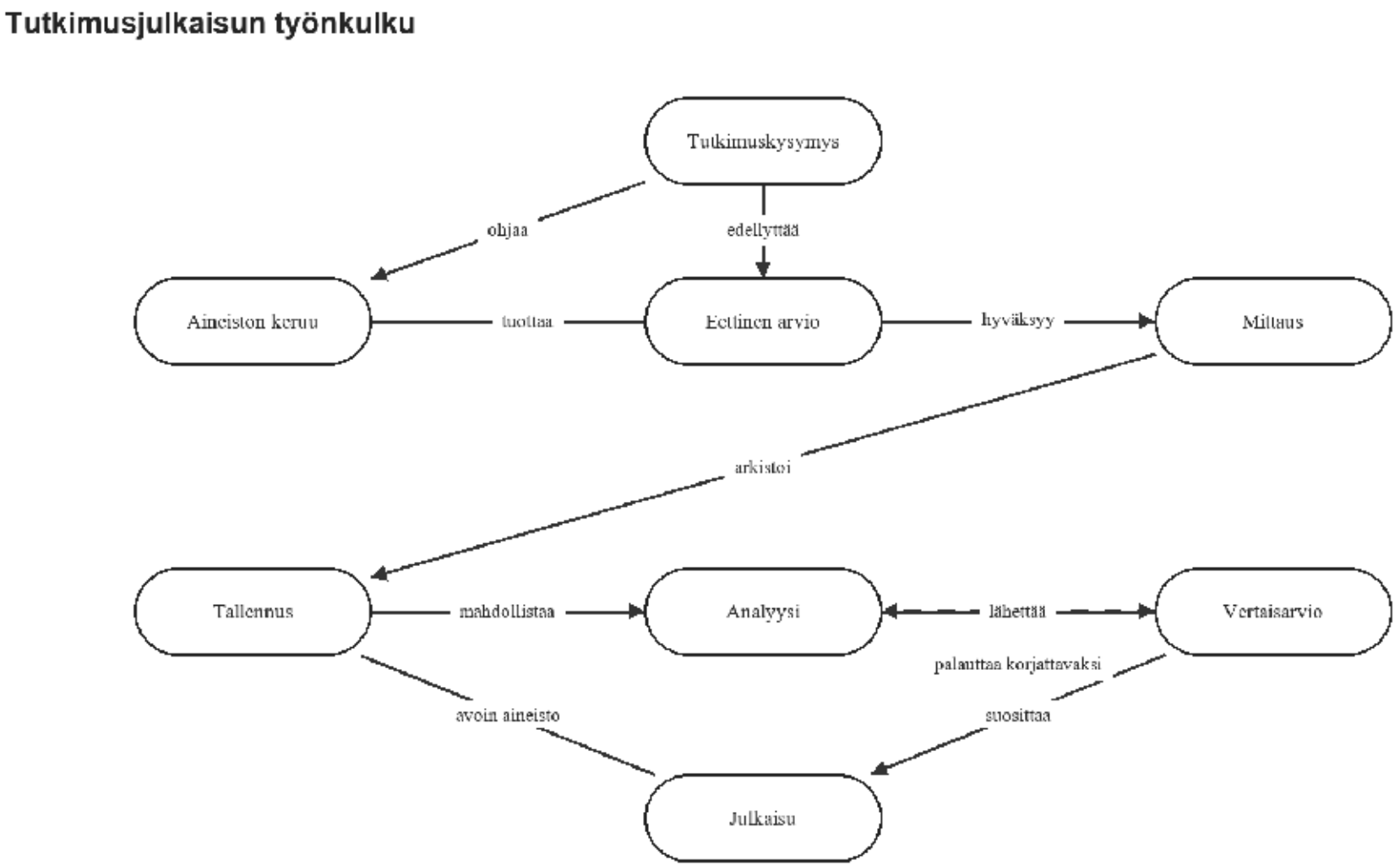
Interpretation

This controlled test demonstrates review-free failure on most tested figures. It does not estimate a universal failure rate: the sample is small and synthetic, and the scorer uses position tolerances.

The OCR evidence union is omitted as a duplicate accuracy row: it added no graph nodes or edges by design.

VISUAL EXAMPLE

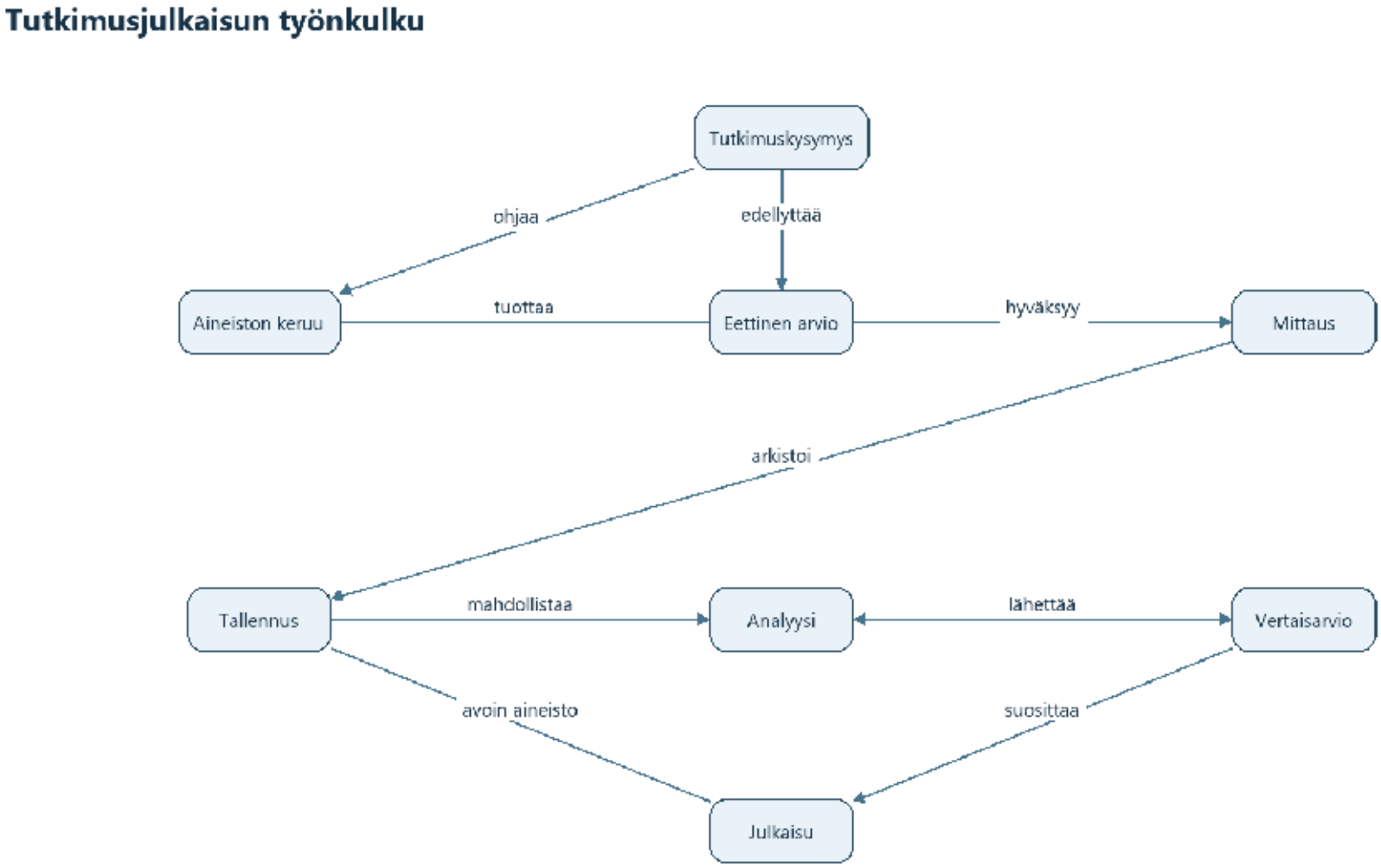
Example source figure



Source graph: 8 nodes and 10 edges, with pill-shaped nodes, arrowless and directed links.

VISUAL EXAMPLE

Blind vision reconstruction



Blind prediction: 8 exact node labels, 8 wrong shapes, 2 missed connections, 1 extra, and 1 wrong arrow.